

ANALYZER

Logic Analyzer for RCBUS using PI PICO 2[W]

PAGE 1 : INTRODUCTION

RcBusInterface



File: RcBusInterface.kicad_sch
PiPicoA



File: PiPicoA.kicad_sch
PiPicoB



File: PiPicoB.kicad_sch
PiPicoC



File: PiPicoC.kicad_sch

PAGE 2 : RCBUS INTERFACE & POWER SUPPLY

PAGE 3 : PI PICO ANALYZER A

PAGE 4 : PI PICO ANALYZER B

PAGE 5 : PI PICO ANALYZER C

BACKGROUND OF DESIGN

This design is a derivative work of John Winans S-100 bus logic analyzer targeted for RCBUS instead of S-100 bus.
<https://github.com/johnwinans/2070-S-100-bus-analyzer>

Both projects use the Dr Gusman software and firmware.
<https://github.com/gusmanb/logicanalyzer>

PCB STACKUP NOTE

JLC04161H-3313 stackup gives :

- Ideal trace impedance (50 & 100ohms).
- GND plane closer to signal layer routing for improved signal integrity.

MOUNTING/TOOLING HOLES

LOG01



LOG02



H4
MountingHole

This documentation describes Open Hardware and is licensed under the CERN OHL v. 1.2.

You may redistribute and modify this documentation under the terms of the CERN OHL v.1.2. (<http://ohwr.org/cernohl>). This documentation is distributed WITHOUT ANY EXPRESS OR IMPLIED WARRANTY, INCLUDING OF MERCHANTABILITY, SATISFACTORY QUALITY AND FITNESS FOR A PARTICULAR PURPOSE. Please see the CERN OHL v.1.2 for applicable conditions.

Logic Analyzer for RCBUS

Title: Logic Analyzer

Size: USLetter

Date: 2026-04-27

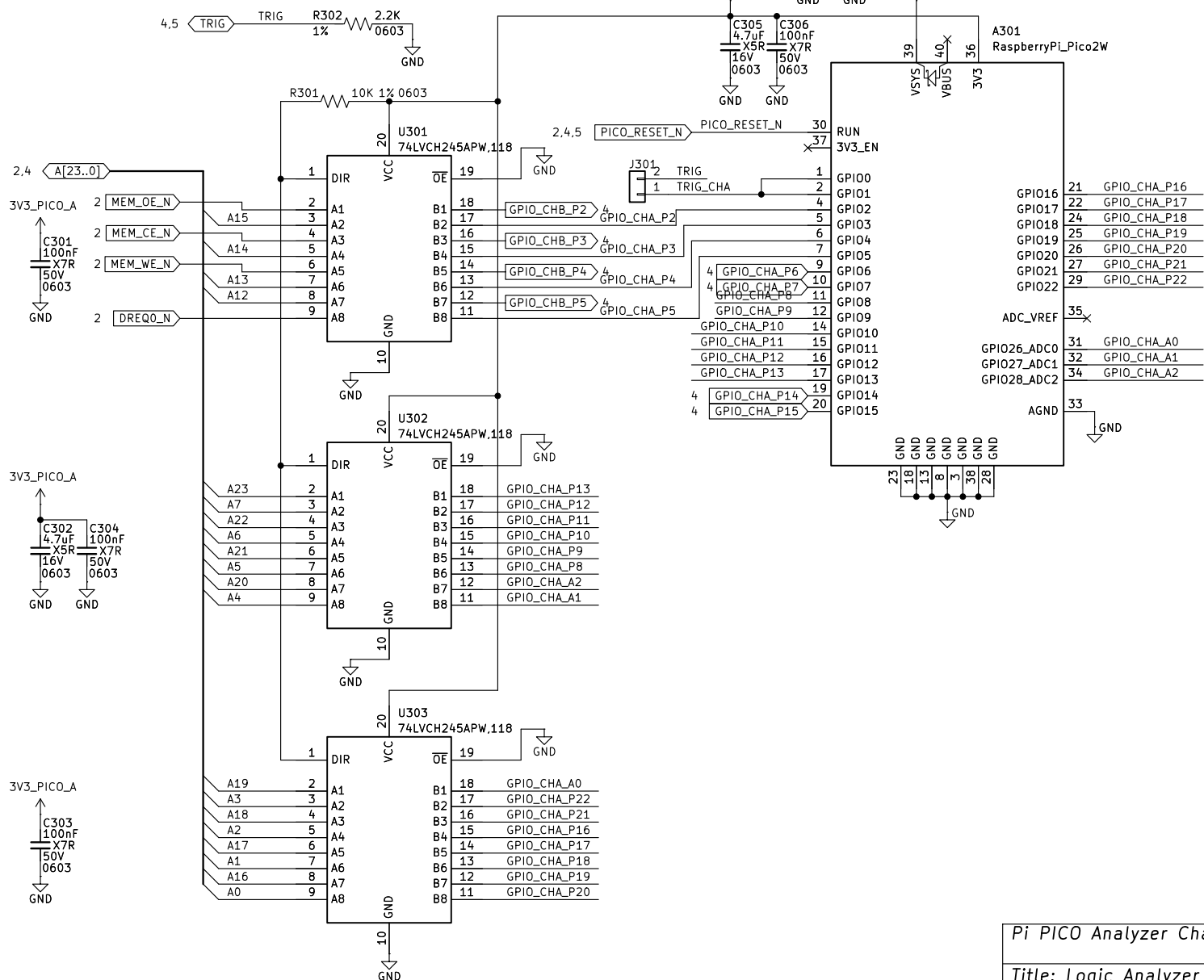
Rev: 0

KiCad E.D.A. 10.0.1

Drawn: Denno Wiggle

Id: 1/5

Pi PICO Channel A



Pi PICO Analyzer Channel A

Title: Logic Analyzer

Size: USLetter

Date: 2026-04-27

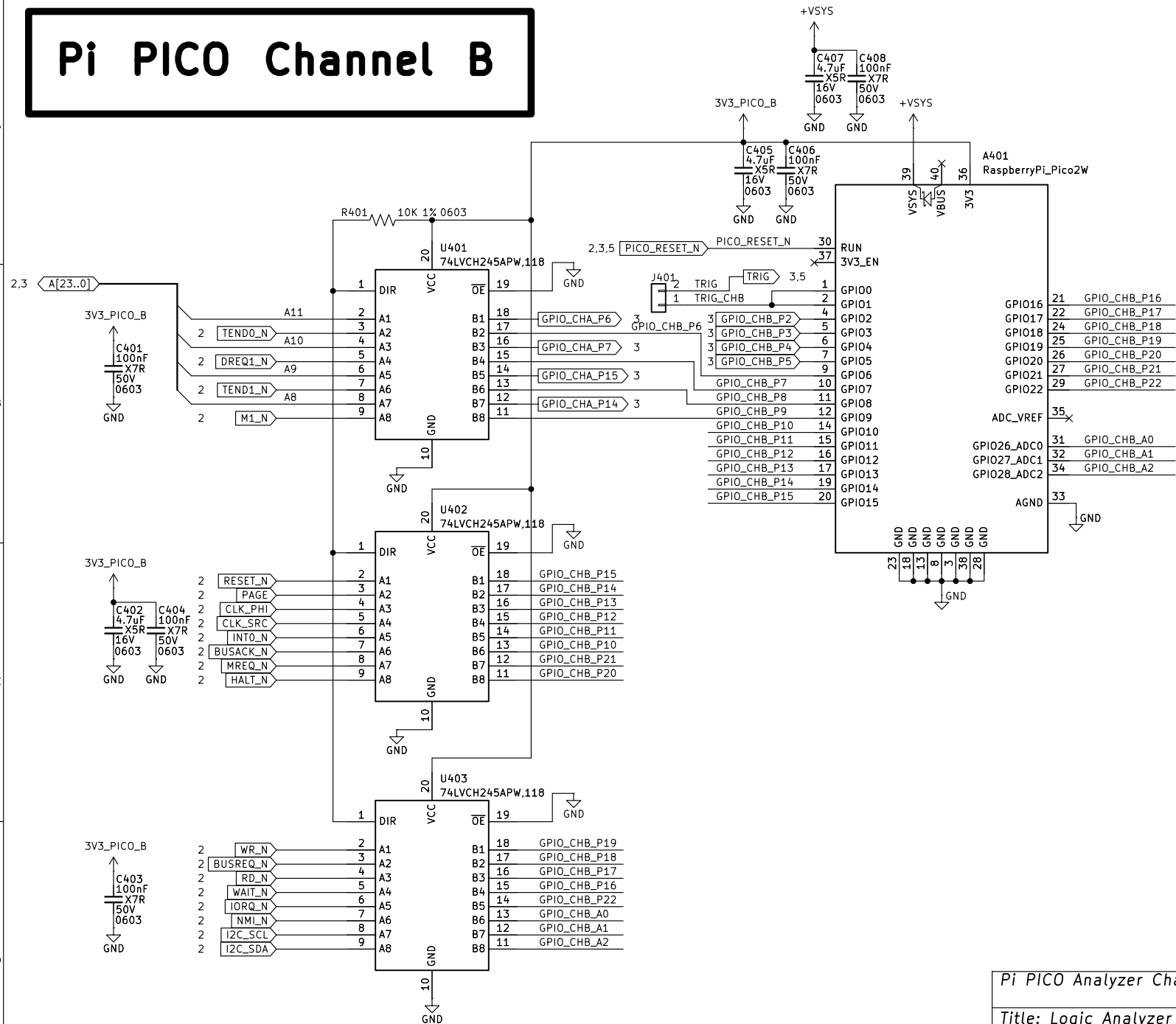
Rev: 0

KiCad E.D.A. 10.0.1

Drawn: Denno Wiggle

Id: 3/5

Pi PICO Channel B



Pi PICO Analyzer Channel B

Title: Logic Analyzer

Size: USLetter

Date: 2026-04-27

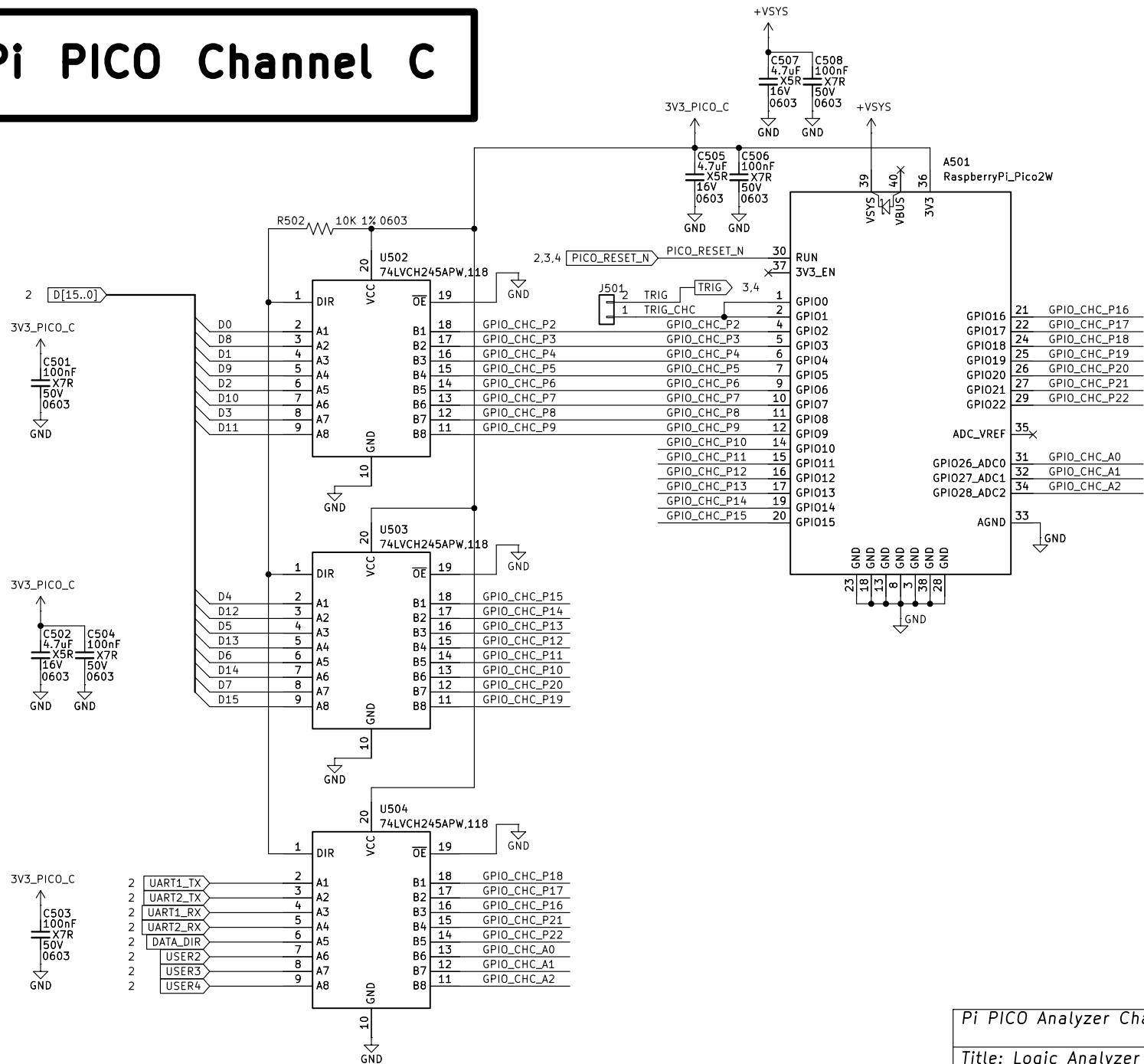
Rev: 0

KiCad E.D.A. 10.0.1

Drawn: Denno Wiggle

Id: 4/5

Pi PICO Channel C



Pi PICO Analyzer Channel C

Title: Logic Analyzer

Size: USLetter

Date: 2026-04-27

Rev: 0

KiCad E.D.A. 10.0.1

Drawn: Denno Wiggle

Id: 5/5